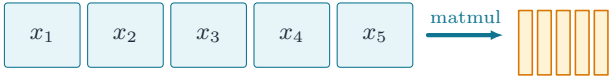
Prefill computes once; decode rereads the whole history every step

PREFILL
whole prompt



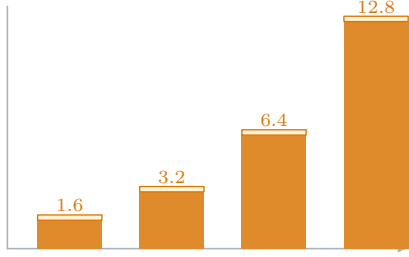
one parallel pass over the whole prompt ⇒ compute-bound, Tensor Cores busy
a cached prefix skips prefill for those positions

the growing read: $2LTn_{kv}d_hb$ bytes per step

DECODE
one query step

append the new K, V: $T \leftarrow T + 1$

$2LTn_{kv}d_hb$ bytes read per step, $\propto T$ ⇒ bandwidth-bound

decode arithmetic intensity $2/b \approx 1$ operation/byte $\ll$ machine balance ≈ 300

KV read time per decode step
orange = HBM read, amber = compute
compute share < 1% of each bar